

Linear Algebra

Bases & Dimension

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01

Introduction

Price Problem



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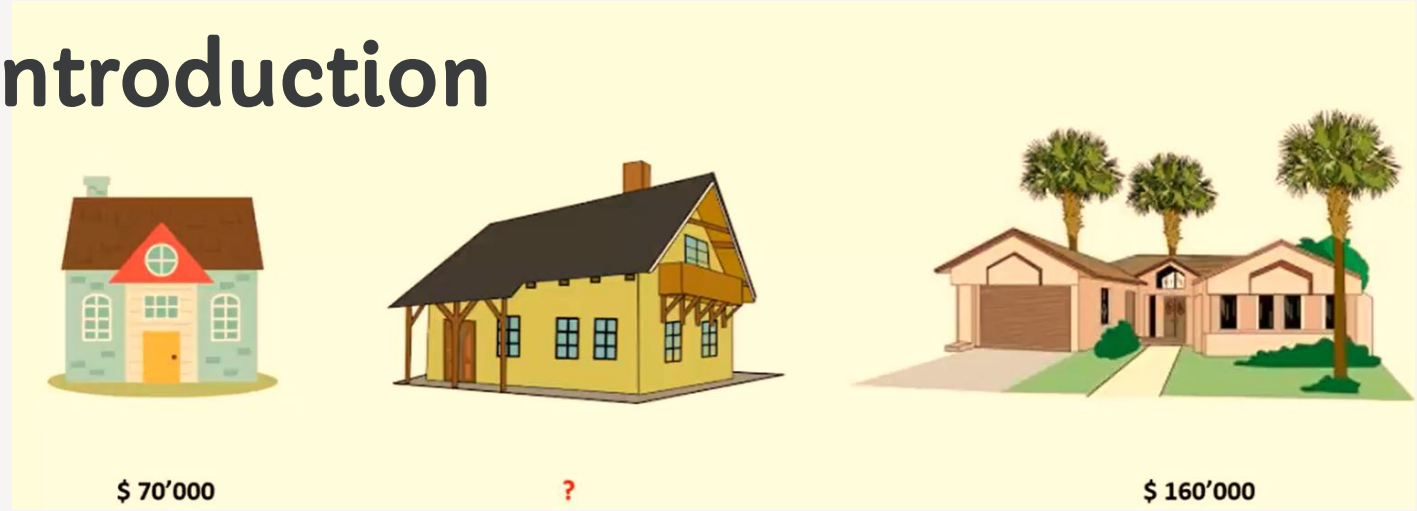


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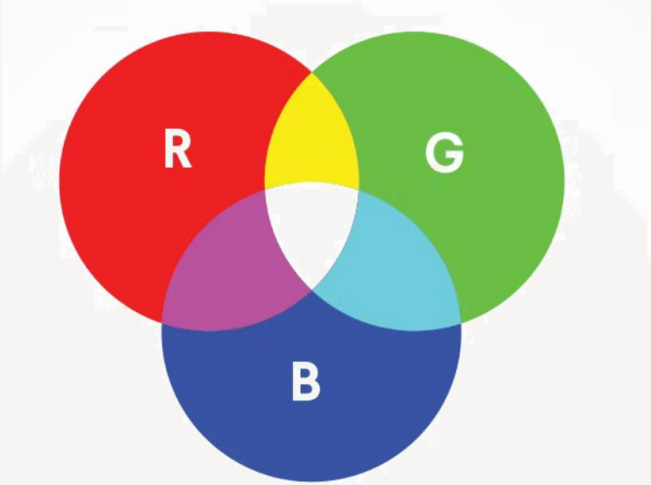
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Introduction



$$
\begin{array}{l}
\\
\text{Home \#1} \\
\text{Home \#2} \\
\text{Home \#3} \\
\\
\text{Home \#N}
\end{array}
\begin{bmatrix}
\#Room & Size_part1 & Size_part2 & Size_part3 & Size_part4 & Size & Age & Floor & Is_near_park \\
 & & & & & & & & \\
 & & & & & & & & \\
 & & & & & & & & \\
 & & & & & & & & \\
 & & & & & & & &
\end{bmatrix}
$$

Introduction





02

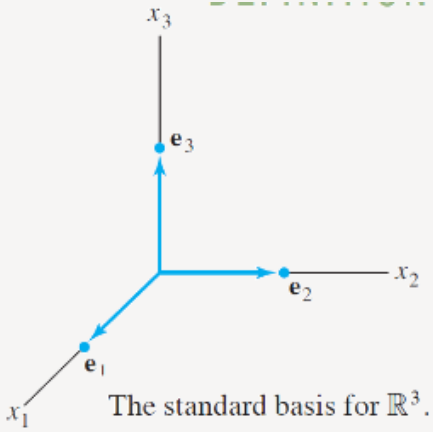
Basis

Basis

- A set of n linearly independent n-vectors is called a basis.
- A basis is the combination of span and independence: A set of vectors $\{v_1, \ldots, v_n\}$ forms a basis for some subspace of $\mathbb{R}^n$ if it

  (1) spans that subspace

  (2) is an independent set of vectors.



The standard basis for $\mathbb{R}^3$.

Basis

Definition

Let H be a subspace of a vector space V. An indexed set of vectors $\mathcal{B} = \{b_1, \dots, b_n\}$ in V is a **basis** for H if

1. $\mathcal{B}$ is linearly independent set, and
2. The subspace spanned by $\mathcal{B}$ coincides with H; that is,

$$H = Span\,\{b_1, \dots, b_n\}$$

Example

Which are unique?

- ❑ Express a vector in terms of any particular basis
- ❑ Bases for $\mathbb{R}^2$
- ❑ Bases with unit length for $\mathbb{R}^2$

Example Solution!

- **Express a vector in terms of any particular basis**
  Unique: Once a basis is fixed, the coefficients (coordinates) are uniquely determined.

- **Bases for $\mathbb{R}^2$**
  Not unique: There are infinitely many possible bases, as any two linearly independent vectors can form one.

- **Bases with unit length for $\mathbb{R}^2$**
  Not unique: There are infinitely many orthonormal bases (rotations of each other).

Vector Space of Polynomials

Be careful: A vector space can have many bases that look very different from each other!

Example (Basis)

- Standard bases for $P_n(\mathbb{R})$?
- Are $(1-x), (1+x), x^2$ basis for $P_2(\mathbb{R})$?

Example Solution!

- **Standard basis for $P_n(\mathbb{R})$:**
  The standard basis is $\{1, x, x^2, \dots, x^n\}$.
  Each polynomial in $P_n(\mathbb{R})$ can be uniquely written as a linear combination of these monomials.

- **Are (1−x), (1+x), x² a basis for $P_2(\mathbb{R})$?**
  Yes. They are **linearly independent**.
  $P_2(\mathbb{R})$ is **spanned** by these three vectors.
  Therefore, they form a valid basis for $P_2(\mathbb{R})$.

03

Dimension

Dimensions

- The dimensionality of a vector is the number of coordinate axes in which that vector exists.
- If a vector space is spanned by a finite number of vectors, it is said to be finite-dimensional. Otherwise, it is infinite-dimensional.
- The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted dim(V).

Bases and finite dimension

Theorem 1

Let V be a vector space which is spanned by a finite set of vectors $x_1, x_2, \dots, x_m$. Then any independent set of vectors in V is finite and contains no more than m elements. Independent ≤ spanning

Conclusion

Every basis of V is finite and contains no more than m elements.

Proof of Theorem 1

Because $\{x_1, \dots, x_m\}$ spans V, every vector in V can be written as a linear combination of these m vectors.

Now suppose we have a linearly independent set $\{y_1, \dots, y_n\}$ in V with $n > m$.

Each y_i can be expressed as

$$y_i = a_{i1}x_1 + \cdots + a_{im}x_m$$

Consider a linear combination of the y_i's that equals the zero vector:

$$0 = c_1y_1 + \cdots + c_ny_n$$

Substitute each y_i in terms of the x_j's:

$$(c_1a_{11} + \cdots + c_na_{n1})x_1 + \cdots + (c_1a_{1m} + \cdots + c_na_{nm})\,x_m = 0$$

Since the x_j's span V, this gives a homogeneous system of m equations in n unknowns $c_1, \dots, c_n$. If $n > m$, this system has a nontrivial solution (some $c_i \neq 0$), which means the y_i's are linearly dependent. ⇒ Contradiction!

Bases and finite dimension

Theorem 2

If V is a finite-dimensional vector space, then any two bases of V has the same (finite) number of elements.

Proof of Theorem 2

Let $B_1 = \{x_1, \dots, x_m\}$ and $B_2 = \{y_1, \dots, y_n\}$ be two bases of V.
Since B_1 spans V, any independent set (like B_2) can have **no more than** m elements $\Rightarrow n \leq m$.
Likewise, since B_2 spans V, any independent set (like B_1) can have **no more than** n elements $\Rightarrow m \leq n$.
Hence $m = n$.

Basis and finite dimension

The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted as dim(V).

Theorem 2



Theorem 3

Let V be a vector space with a basis B of size m. Then

a) Any set of more than m vectors in V must be linearly dependent, and

b) Any set of fewer than m vectors cannot span V.

Proof of Theorem 3

a) Since B is a basis, it spans V. By theorem 2, any linearly independent set in V can have at most m elements. Therefore, any set with more than m vectors must be linearly dependent.

b) Suppose, for contradiction, that there exists a spanning set $\{y_1, \dots, y_k\}$ with $k < m$. From S, we can extract a basis (by removing dependent vectors). That basis would have at most k elements. But this contradicts the fact that every basis of V must have m elements.

Dimensions

Definition

A vector space V is called...

a) **finite-dimensional** if it has a finite basis, and its **dimension**, denoted by $\dim(V)$, is the number of vectors in one of its bases.

b) **infinite-dimensional** if it has no finite basis, and we say that $\dim(V) = \infty$.

Note

Dimension of subspace $\{\mathbf{0}\}$ is zero.

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Finite Dimensional Subspace

Basis of Subspace

Theorem 4

If W is a subspace of a finite-dimensional vector space V, every linearly independent subset of W is finite and is part of a (finite) basis for W.

Lemma

Let S be a linearly independent subset of a vector space V. Suppose u is a vector in V which is not in the subspace spanned by S. Then the set obtained by adjoining u to S is linearly independent.

Proof of Lemma

Assume

$$a_1v_1 + \cdots + a_nv_n + cu = 0$$

If $c \neq 0$, we can solve for u: $u = -\frac{1}{c}(a_1v_1 + \cdots + a_nv_n)$

This means u is a linear combination of vectors in S, i.e., $u \in \text{span}(S)$, which contradicts our assumption that $u \notin \text{span}(S)$.

Therefore, $c = 0$. The equation becomes $a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0$.

Since S is linearly independent, this implies $a_1 = a_2 = \cdots = a_n = 0$.

Thus, all coefficients are zero, so $S \cup \{u\}$ is linearly independent.

Proof of Theorem 4

Let $\dim V = m$. Every linearly independent set in V has at most m elements.

Hence any linearly independent subset of W also has size $\leq m$, so it is finite.

Therefore, the possible sizes of independent subsets of W lie in the finite set $\{0,1,\dots,m\}$.

So largest possible size exists. Choose an independent subset $B \subseteq W$ that has this maximal size.

We claim B spans W. If not there exits a $w \notin span(B)$.

By lemma, $B \cup \{w\}$ must be linearly independent, contradicting maximality of B. Thus, B is a finite basis for W.

Basis of Subspace

Corollary

A subspace is called a proper subspace **if it's not the entire space**, so R2 is the only subspace of R2 which is not a proper subspace

If W is a proper subspace of a finite-dimensional vector space V, then W is finite-dimensional and $\dim(W) < \dim(V)$

Corollary

In a finite-dimensional vector space V, every non-empty linearly independent set of vectors is part of basis.

Basis of sum of subspaces

Theorem 5

If W_1 and W_2 are finite-dimensional subspaces of a vector space V,

the $W_1 + W_2$ is a finite-dimensional and

$$\dim(W_1) + \dim(W_2) = \dim(W_1 \cap W_2) + \dim(W_1 + W_2)$$

Proof of Theorem 5

Let $v_1, \dots, v_m$ be a basis of $W_1 \cap W_2$, so $\dim(W_1 \cap W_2) = m$.

Extend this basis to a basis of W_1 : $v_1, \dots, v_m, u_1, \dots, u_j$,so $\dim(W_1) = m + j$.

Similarly, extend $v_1, \dots, v_m$ to a basis of W_2: $v_1, \dots, v_m, w_1, \dots, w_k$,so $\dim(W_2) = m + k$.

We will show that $v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$ is a basis of $W_1 + W_2$.

1. Spanning:

Every vector in W_1 is a linear combination of v's and u's, and every vector in W_2 is a linear combination of v's and w's. Hence the span of all these vectors contains both W_1 and W_2, so it equals $W_1 + W_2$.

Proof of Theorem 5 (cont.)

2. Linear Independence:

Suppose $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j + c_1w_1 + \cdots + c_kw_k = 0$.

Move the w-terms to one side:

$-(a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j) = c_1w_1 + \cdots + c_kw_k$.

The right-hand side is in W_1, and the left-hand side is in W_2;hence this vector is in $W_1 \cap W_2$, which is spanned by $v_1, \ldots, v_m$ .Thus

$d_1v_1 + \cdots + d_mv_m = c_1w_1 + \cdots + c_kw_k$ for some d_i's.

Since $v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k$ are linearly independent all c_i's (and d_i's) are 0, so we have: $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0$ which implies all a_i's and b_i's 0 since these vectors are basis of W_1. Hence all vectors are linearly independent.

Basis of sum of subspaces

Theorem 6

If $W_1,\ W_2\ and\ W_3$ are finite-dimensional subspaces of a vector space V, then can we have the following relation?

$$\dim(W_1 + W_2 + W_3)$$
$$= \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2)$$
$$- \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3)$$

Counterexample: $W_1 = span\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}, W_2 = span\{\begin{bmatrix}0\\1\end{bmatrix}\}, W_3 = span\{\begin{bmatrix}1\\1\end{bmatrix}\}$

Basis of sum of subspaces

Theorem 6

If $W_1,\ W_2\ and\ W_3$ are finite-dimensional subspaces of a vector space V, then:

$$\dim(W_1 + W_2 + W_3)$$
$$\leq \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2)$$
$$- \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3)$$

Problem 24: *Let W_1, W_2, W_3 be subspaces of a F.D.V.S. Show that*

$$\dim (W_1 + W_2 + W_3) \leq \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_1 \cap W_2) - \dim (W_1 \cap W_3) - \dim (W_2 \cap W_3) + \dim (W_1 \cap W_2 \cap W_3).$$

Solution: We have

$$\begin{aligned}\dim (W_1 + W_2 + W_3) &= \dim W_1 + \dim (W_2 + W_3) - \dim (W_1 \cap (W_2 + W_3)) \\ &= \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_2 \cap W_3) \\ &\qquad - \dim (W_1 \cap (W_2 + W_3)) \\ &\leq \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_2 \cap W_3) \\ &\quad - \dim (W_1 \cap W_2) - \dim (W_1 \cap W_3) + \dim (W_1 \cap W_2 \cap W_3)\end{aligned}$$

as $(W_1 \cap W_2) + (W_1 \cap W_3) \subseteq W_1 \cap (W_2 + W_3)$.

Dimensionality and Properties of Bases

Note

Let V be a finite dimensional vector space over field F. Below are some properties of bases:

1. Any linearly independent list can be extended to a basis (a maximal linearly independent list is spanning).
2. Any spanning list contains a basis (a minimal spanning list is linearly independent).
3. Any linearly independent list of length dim V is a basis.
4. Any spanning list of length dim V is a basis.

We will learn about change of basis later.

05

Coordinates

Ordered basis

Definition

If V is a finite-dimensional vector space, as ordered basis for V is a finite sequence of vectors which is linearly independent and spaces V.

Be careful: The order in which the basis vectors appear in B affects the order of the entries in the coordinate vector. This is kind of janky (technically, sets don't care about order), but everyone just sort of accepts it.

Coordinate Systems

- The main reason for selecting a basis for a subspace H; instead of merely a spanning set, is that each vector in H can be written in only one way as a linear combination of the basis vectors.

Note

Suppose the set $\mathcal{B} = \{\boldsymbol{b_1}, \dots, \boldsymbol{b_P}\}$ is a basis for a subspace H. For each x in H, the **coordinates of x relative to the basis $\mathcal{B}$** are the weights $c_1, \dots, c_P$ such that $\mathrm{x} = c_1 b_1 + \cdots + c_P b_p$, and the vector in $\mathbb{R}^p$

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_P \end{bmatrix}$$

is called the **coordinate vector of x (relative to $\mathcal{B}$)** or the $\mathcal{B}$-coordinate vector of x.

Coordinate Systems

Example

Coordinate vector of $p(x) = 4 - x + 3x^2$ respect to basis $\{1, x, x^2\}$

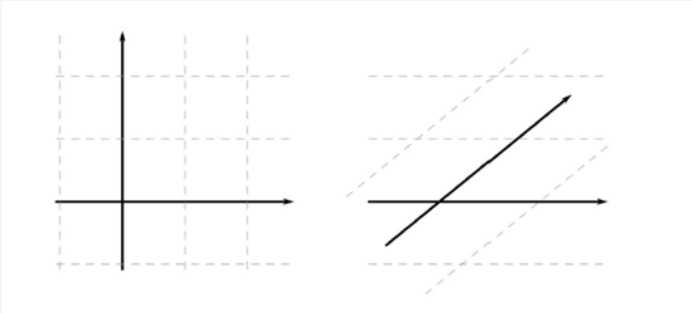
Example Solution!

$$p(x) = 4(1) - 1(x) + 3(x^2)$$

$$[p(x)]_B = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$$

Coordinate axes

- The familiar Cartesian plane (left) has orthogonal coordinate axes. However, axes in linear algebra are not constrained to be orthogonal (right), and non-orthogonal axes can be advantageous.



Resources

- Page 97 LINEAR ALGEBRA: Theory, Intuition, Code
- Page 213: David Cherney,
- Page 54: Linear Algebra and Optimization for Machine Learning

Dimensions

Example

Let's compute the dimension of some vector spaces that we've been working with.

Vector space	Basis	Dimension
$\mathcal{F}^n$ (n–tuples each elements from field $\mathcal{F}$)		
P^p (polynomials with max degree p)		
$M_{m,n}$ (matrices with m rows and n columns)		
P (all polynomials)		
F $(all\ functions)$		
C (all continues functions)		

Note!

Example. *Let's compute the dimension of some vector spaces that we've been working with.*

V	Basis	$\dim(V)$
$\mathbb{F}^n$ ($\mathbb{R}^n$, $\mathbb{C}^n$, etc.)	$\{\vec{e}_1, \vec{e}_2, \cdots, \vec{e}_n\}$	n
P^p	$\{1, x, x^2, \cdots, x^p\}$	$p+1$ ← not p!
$M_{m,n}$	$\{E_{1,1}, E_{1,2}, \cdots, E_{m,n}\}$	mn
P (all polynomials)	$\{1, x, x^2, x^3, \cdots\}$	∞
$\mathcal{F}$	??	∞
$\mathcal{C}$ (continuous functions)	??	∞